BEMS GROUP ENTERPRISE ERP

AI Automation & n8n Integration Guide

Comprehensive Manual for Autonomous Corporate Operations, Payroll, Attendance & Staged Approvals

Document Version: 1.1.0 | **Environment:** Production (https://bemserp.work) | **Target Audience:** AI Automation Engineer

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EXECUTIVE NOTICE & SCOPE BOUNDARIES:
• **Strict 3-Role Governance:** Only SUPER_ADMIN, GMD, and STAFF exist in the entire system. Never invent or pass legacy roles.
• **Approval Staging Guarantee:** The AI *never* approves expenditures directly. It stages actions and dispatches interactive confirmation cards with banking app deep links.
• **File Upload Exclusion:** Physical file uploads (logos, certificates, invoices, photo receipts) are **excluded** from AI automation tools. All files are uploaded normally via the Web Dashboard.
• **Zero Dummy Data Rule:** AI workflows must collect exact, verified parameters from the GMD before invoking creation endpoints. Incomplete requests return HTTP 400.

1. Authentication & Global Headers

All requests made by n8n or external automation scripts must authenticate against the production server using the Master AI Service Bearer Token:

Header Name	Header Value	Description
`Authorization`	`Bearer 7095ec2348ee3a4ca855707cec7bab8a697f0de4aad3f93f0e545d82bba8e9a8`	Master AI Service Token
`x-api-key`	`7095ec2348ee3a4ca855707cec7bab8a697f0de4aad3f93f0e545d82bba8e9a8`	Alternate header format
`Content-Type`	`application/json`	Mandatory for POST requests

2. Complete AI Tool API Endpoints Directory

Domain / Feature	Method & URL	Action / Capability
Service Discovery	`GET /api/ai/tools`	Lists all available AI tools & capability descriptors
Corporate Subsidiaries	`GET /api/ai/tools/subsidiary` `POST /api/ai/tools/subsidiary`	List operating entities or provision a new subsidiary with mandatory parameters
Staged Approvals & Payouts	`GET /api/ai/tools/approvals` `POST /api/ai/tools/approvals`	Fetch pending approvals; stage approval with deep links into banking apps & USSD
Bank Accounts	`GET /api/ai/tools/bank-accounts` `POST /api/ai/tools/bank-accounts`	Look up or register bank accounts for STAFF, SUBSIDIARY, or VENDOR with NUBAN check
Executive Onboarding	`POST /api/ai/tools/admin-onboarding`	Onboard SUPER_ADMIN or GMD via cryptographic invitation email dispatch
Full Payroll Operations	`GET /api/ai/tools/payroll` `POST /api/ai/tools/payroll`	Run payroll calculation with attendance deductions; lock batch; mark paid
Attendance & Timekeeping	`GET /api/ai/tools/attendance` `POST /api/ai/tools/attendance`	Inspect daily master roster; excuse penalty; clock in/out; update shift policy
Vendor & 3-Way Match	`GET /api/ai/tools/vendors` `POST /api/ai/tools/vendors`	Master vendor registry, NUBAN check, and 3-way match verification (PO vs GRN vs Invoice)
Intercompany Ledger	`GET /api/ai/tools/intercompany` `POST /api/ai/tools/intercompany`	Automated paired Debit/Credit notes, cross-entity matrix, and GMD netting settlement
Conversational Chatbot	`POST /api/ai/chat`	Direct natural language command execution & checklist generator

3. Corporate Subsidiary Provisioning

Endpoint: POST https://bemserp.work/api/ai/tools/subsidiary
Rule: File uploads are excluded from AI tools. Upload logos and legal certificates on the Web Dashboard.

```
{
  "name": "Arella Creative Studio Ltd",
  "brandColor": "#EA580C",
  "industry": "Media & Creative Production",
  "headquarters": "14 Admiralty Way, Lekki Phase 1, Lagos",
  "code": "ACS",
  "currency": "NGN"
}
```

4. Staged Approvals & Quick-Pay Bank Deep Links

Endpoint: POST https://bemserp.work/api/ai/tools/approvals
Protocol: The AI *never* executes direct approval. It creates a staged action and returns deep links into Nigerian banking apps (GTWorld, Kuda, Zenith, Access, OPay, Moniepoint) and USSD strings for fast GMD settlement.

```
{
  "requestId": "REQ-2026-0005",
  "decision": "APPROVE",
  "comment": "Approved for immediate deployment"
}
```

Response Output Structure:
• status: 'AWAITING_CONFIRMATION'
• beneficiaryAccount: { bankName, accountNumber, accountName }
• links.quickPayPortal: 'https://bemserp.work/pay/REQ-2026-0005'
• links.fastUssdDialer: '*737*2*150000*2001928374#'
• stagedAction.confirmationEndpoint: '/api/ai/staged-actions/{id}/confirm'

5. Bank Account Registration & Verification

Endpoint: POST https://bemserp.work/api/ai/tools/bank-accounts
Registers verified 10-digit NUBAN accounts for staff members, subsidiaries, or vendors.

```
{
  "accountType": "STAFF",
  "targetQuery": "babatunde@bemsgroup.com",
  "bankName": "Kuda Microfinance Bank",
  "accountNumber": "2001928374",
  "accountName": "Babatunde Diekolayomi"
}
```

6. Executive Admin Onboarding (Strict 3-Role Rule)

Endpoint: POST https://bemserp.work/api/ai/tools/admin-onboarding
Strict Rule: Only SUPER_ADMIN or GMD can be invited. Staff members register exclusively via self-service mobile app. All other roles are rejected.

```
{
  "legalName": "Engr. Michael Adeyemi",
  "workEmail": "adeyemi@bemsgroup.com",
  "phone": "+2348030001122",
  "role": "SUPER_ADMIN"
}
```

7. Full Payroll Operations Suite

Endpoint: POST https://bemserp.work/api/ai/tools/payroll
Automates attendance penalty deductions (5% lateness, 100% absence) and generates payslip items.

```
// Action 1: Calculate & Prepare Payroll
{ "action": "PREPARE", "month": 11, "year": 2026 }

// Action 2: Sign Off & Lock Period
{ "action": "SIGN_OFF", "batchId": "BATCH-2026-11" }

// Action 3: Record Payouts Disbursed
{ "action": "RECORD_PAYOUT", "batchId": "BATCH-2026-11", "reference": "GTB-BULK-928371" }
```

8. Attendance Operations & Penalty Waivers

Endpoint: POST https://bemserp.work/api/ai/tools/attendance

```
// Action 1: Waive / Excuse Lateness Penalty
{ "action": "EXCUSE_OVERRIDE", "employeeId": "emp-uuid", "date": "2026-11-05", "reason": "Official field assi
gnment approved by GMD" }

// Action 2: Update Shift & Resumption Policy
{ "action": "UPDATE_POLICY", "workStartTime": "08:00", "gracePeriodMinutes": 15, "latenessDeductionPercent":
5.0 }
```

8.1. Vendor Registry & Automated 3-Way Match Verification

Endpoint: POST https://bemserp.work/api/ai/tools/vendors
Verifies Purchase Order against Goods Receipt Note (GRN) acceptance and Vendor Invoice billed amount. 0% tolerance on unapproved overage.

```
// Action 1: Check 3-Way Match Status on PO
{ "action": "CHECK_MATCH", "requestId": "REQ-2026-0005" }

// Action 2: Register Verified Vendor with NUBAN Bank Account
{
  "action": "REGISTER",
  "name": "Mikano International Ltd",
  "category": "FACILITIES",
  "bankName": "Guaranty Trust Bank",
  "accountNumber": "0123456789",
  "accountName": "Mikano International Ltd"
}
```

8.2. Automated Intercompany Debit/Credit Ledger & Netting

Endpoint: POST https://bemserp.work/api/ai/tools/intercompany
Generates paired Debit/Credit notes when one entity pays on behalf of another entity. Stages netting settlements for GMD confirmation.

```
// Action 1: Generate Paired Operational Debit/Credit Notes
{
  "action": "CREATE_NOTE",
  "source": "BEMS-GROUP",
  "target": "ACS",
  "amount": 350000,
  "description": "Cross-charge: Q4 Starlink Equipment Allocation"
}

// Action 2: Stage Netting Settlement for GMD Authorization
{ "action": "STAGE_SETTLEMENT", "noteIds": ["note-uuid-1", "note-uuid-2"] }
```

9. n8n Node Blueprint & WhatsApp Notification Template

WhatsApp / Slack Message Template for Dispatched Approvals:

```
■ *REQ-2026-0005 APPROVED*
*Title:* Starlink Router Replacement
*Amount:* ■150,000
*Beneficiary:* Babatunde Diekolayomi
*Bank:* Kuda Microfinance Bank
*Account:* 2001928374

■ *Tap to Disburse in Bank App:*
https://bemserp.work/pay/REQ-2026-0005

■ *Fast Phone USSD:*
*737*2*150000*2001928374#
```

10. Mandatory Parameter Checklist Protocol

Before sending POST requests, n8n LLM nodes must ask the GMD for all mandatory specifications:

- **Subsidiary:** Legal Corporate Name, Brand Color (Hex), Business Sector, Headquarters Location.
- **Executive Admin:** Full Legal Name, Corporate Work Email, Mobile Phone, Role (strictly SUPER_ADMIN or GMD).
- **Bank Account:** Target Type (STAFF/SUBSIDIARY/VENDOR), Target Identifier, Bank Name, 10-Digit NUBAN, Verified Account Name.
- **Payroll:** Action (PREPARE/SIGN_OFF/RECORD_PAYOUT), Month, Year.

11. System Scope & Boundary Guarantees

1. **File Upload Exclusion:** Physical file uploads (logos, certificates, invoices, photo receipts) are excluded from AI tools and handled directly on the Web Dashboard.
2. **Approval Staging Guarantee:** The AI never executes direct fund approval; it stages the action and dispatches deep links and interactive buttons for the GMD.
3. **Strict 3-Role Architecture:** Only SUPER_ADMIN, GMD, and STAFF exist in the system. Never introduce legacy roles.

ISSUED BY: BEMS Group Executive Management & Engineering Team
STATUS: Active & Enforced in Production (bemserp.work)
SECURITY: Authorized Engineering Automation Integration Only